StudySapiens Solved Notes

Ch.5 Enzymes

Biology 11 • Punjab English Medium • PECTAA 2025–26 • Solved Exercise
Prompts are paraphrased and keyed to textbook question numbers. MCQ answer letters refer to the textbook options.

SECTION 1: MULTIPLE CHOICE QUESTIONS

Answer letters refer to the textbook options. Full textbook stems/options are not reproduced here.

1. What role does NAD play in oxidative pathways?

Answer: (b) Coenzyme — NAD^+ is an organic coenzyme that carries electrons/hydrogen in oxidation-reduction reactions.

2. Enzymes that join two molecules using ATP belong to which class?

Answer: (b) Ligase — Ligases form new bonds between molecules using energy from ATP.

3. Which is an example of a hydrolase?

Answer: (a) Lipase — Lipase hydrolyses lipids into fatty acids and glycerol.

4. Which statement about enzymes is correct?

Answer: (c) They are specific for the reactions they catalyse — Enzymes lower activation energy, are not consumed, and show substrate/reaction specificity.

5. Zn^{2+} required by an enzyme is best identified as what?

Answer: (b) Activator — Inorganic ions such as Zn^{2+} act as cofactors/activators that help enzyme catalysis.

6. If an enzyme solution is saturated with substrate, how can product formation be increased fastest?

Answer: (a) Add more enzyme — At substrate saturation, active sites are occupied; adding enzyme provides more active sites.

7. How does a non-competitive inhibitor reduce enzyme rate?

Answer: (b) By changing the shape of the enzyme — It binds away from the active site and alters enzyme conformation, reducing catalytic activity.

8. Which enzyme class catalyses addition of water to break a substrate?

Answer: (c) Hydrolase — Hydrolases break bonds by adding water (hydrolysis).

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SECTION 2: SHORT QUESTIONS

1. Define enzyme and cofactor.

Enzymes are biological catalysts, usually proteins, that speed up reactions by lowering activation energy without being consumed. A cofactor is a non-protein helper required by some enzymes; it may be an inorganic ion, coenzyme or prosthetic group.

2. Differentiate between coenzyme and prosthetic group.

Coenzyme	Prosthetic group
Binding: Loosely/transiently bound organic molecule.	**Binding:** Tightly/permanently bound non-protein group.
Example: NAD^+, FAD, coenzyme A.	**Example:** Haem in catalase/peroxidase, biotin.
Role: Often carries electrons or chemical groups.	**Role:** Remains attached and participates in catalysis.

3. What are hydrolases? Give two examples.

Hydrolases catalyse hydrolysis, breaking bonds by adding water. Examples: lipase (lipids), amylase (starch), peptidase (proteins).

4. What is activation energy?

Activation energy is the minimum energy needed to start a chemical reaction. Enzymes lower this barrier so reactions proceed faster at body temperature.

5. Define feedback inhibition.

Feedback inhibition is regulation in which the end product of a pathway inhibits an enzyme early in the pathway, preventing overproduction. Example: ATP inhibiting an enzyme of ATP synthesis.

6. Give examples of competitive and non-competitive inhibitors.

Competitive: malonate inhibits succinate dehydrogenase; sulfa drugs compete with PABA. Non-competitive: heavy metal ions such as Hg^{2+}/Pb^{2+}, cyanide, and ATP feedback inhibition in some pathways.

7. What is optimum pH? Give optimum pH of three human enzymes.

Optimum pH is the pH at which an enzyme works fastest. Examples: pepsin ≈ 2, salivary amylase ≈ 6.7–7.0, trypsin ≈ 8.

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SECTION 3: LONG QUESTIONS

1. Describe enzyme structure and the role/component parts of the active site.

Most enzymes are globular proteins with a specific three-dimensional tertiary structure. Some also have quaternary structure with multiple subunits. The folded shape creates a small region called the active site.

- **Binding site:** amino-acid residues that recognize and hold the substrate through shape, charge and hydrophobic/hydrophilic interactions.
- **Catalytic site:** residues that weaken substrate bonds, orient reactants and stabilize the transition state, converting substrate to product.
- **Specificity:** depends on the precise geometry and chemistry of the active site.

After product release, the enzyme is unchanged and can catalyse another reaction.

2. Differentiate among the three types of cofactors with examples.

Activators/inorganic ions	Coenzymes	Prosthetic groups
Nature: metal ions that assist catalysis.	**Nature:** loosely/transiently bound organic molecules.	**Nature:** tightly/permanently bound non-protein groups.
Role: stabilize charges, participate in redox or substrate binding.	**Role:** carry electrons, hydrogen or chemical groups between enzymes.	**Role:** remain attached and participate directly in catalysis.
Examples: Zn^{2+}, Mg^{2+}, Cl^{-}, Fe^{2+}/Fe^{3+}.	**Examples:** NAD^{+}, FAD, coenzyme A.	**Examples:** haem in catalase/peroxidase, biotin, FAD in some enzymes.

Apoenzyme (protein part) + cofactor = holoenzyme (active enzyme).

3. Explain induced-fit model and compare it with lock-and-key model.

Lock-and-key model	Induced-fit model
Active site: rigid and exactly complementary to substrate.	**Active site:** flexible; changes shape on substrate binding.
Binding: substrate fits like a key in a lock.	**Binding:** enzyme moulds around substrate for tighter fit.
Strength: explains specificity simply.	**Strength:** better explains catalysis and transition-state stabilization.

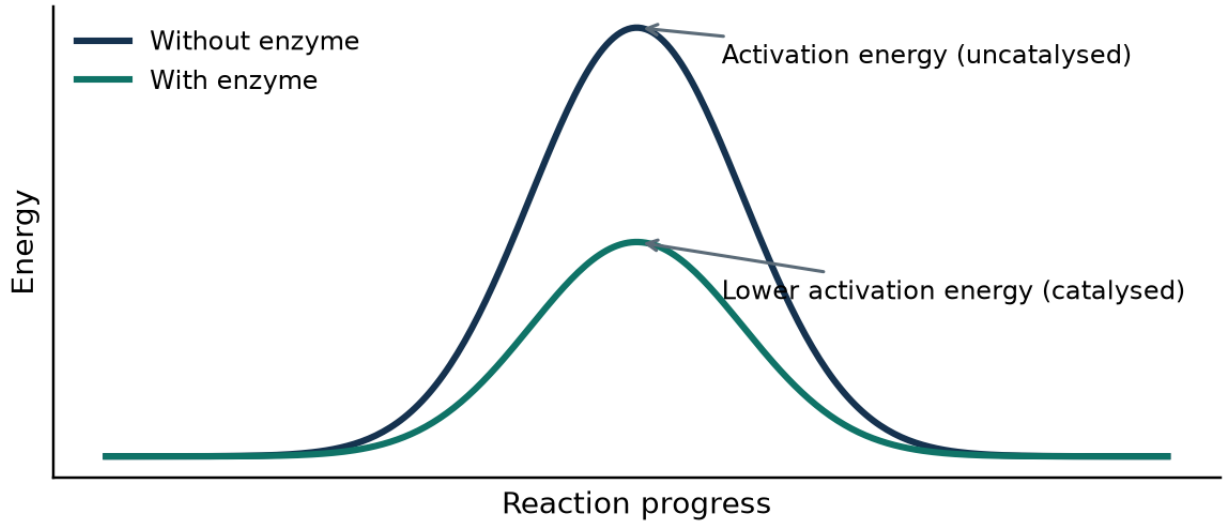
In induced fit, substrate binding causes a conformational change that positions catalytic groups correctly, strains substrate bonds and lowers activation energy.

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This dynamic model is more widely accepted today.

4. Define activation energy and explain with a graph how enzymes speed reactions.

Activation energy is the minimum energy barrier that reactants must cross to reach the transition state and form products. Even exergonic reactions need this initial input.



An enzyme lowers the activation-energy barrier without changing the overall energy difference between reactants and products.

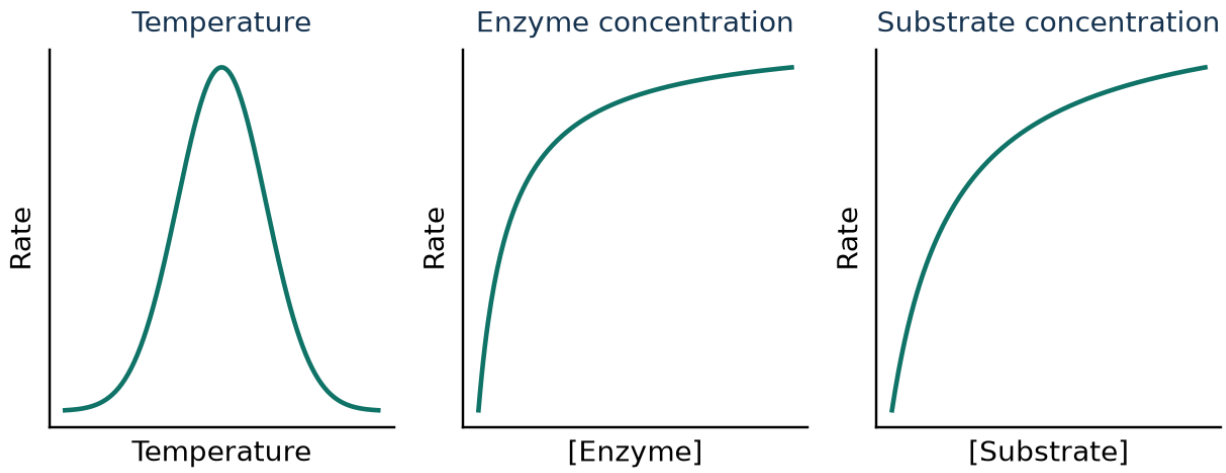
Enzymes provide an alternative pathway with lower activation energy, so a larger fraction of molecules can react per unit time. The equilibrium and overall ΔG remain unchanged; only the rate increases.

5. Describe the effect of temperature on enzyme action.

As temperature rises, molecules move faster and collide more often, so reaction rate increases up to an optimum temperature. For most human enzymes this is near 37°C.

Above the optimum, heat breaks hydrogen and ionic bonds maintaining tertiary structure; the enzyme denatures and the active site loses shape, so rate falls sharply. Low temperature slows molecular motion and reduces activity but usually does not denature the enzyme; activity can return on warming.

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Temperature curve shows an optimum followed by a sharp decline due to denaturation.

6. Compare optimum temperatures of human and thermophilic bacterial enzymes.

Human enzymes usually have optimum temperatures near 37°C, matching normal body temperature. Above about 40–45°C many begin to denature.

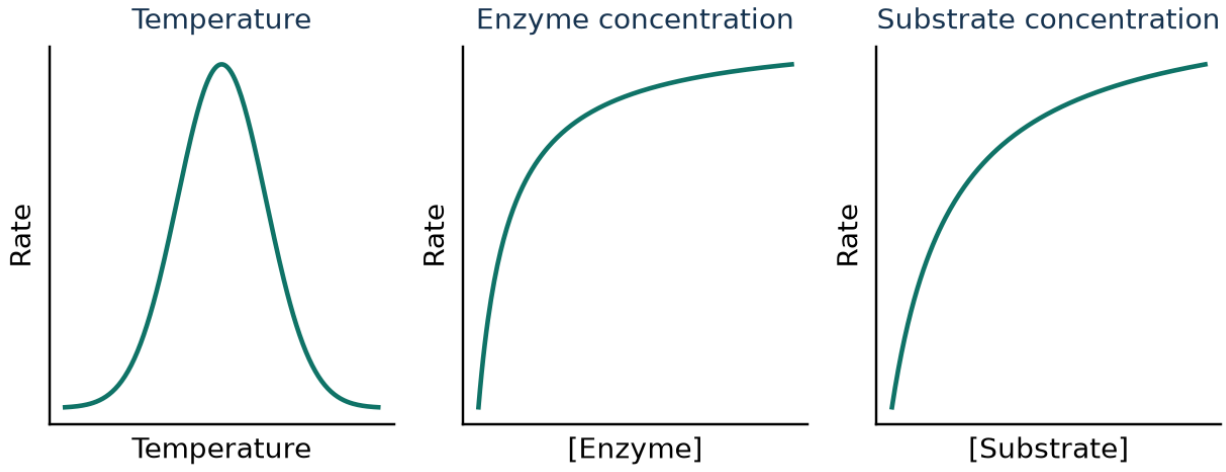
Thermophilic bacteria live in hot springs and hydrothermal vents; their enzymes have optima often at 70°C or above. Their proteins are stabilized by extra ionic bonds, hydrophobic cores and compact folding, so they remain functional at temperatures that would denature human enzymes. Example: Taq polymerase used in PCR.

7. Describe how enzyme concentration affects reaction rate.

With excess substrate, increasing enzyme concentration increases reaction rate almost linearly because more active sites are available to convert substrate to product.

If substrate becomes limiting, adding more enzyme no longer increases rate because all substrate molecules are already being processed. Thus the effect of enzyme concentration depends on substrate availability.

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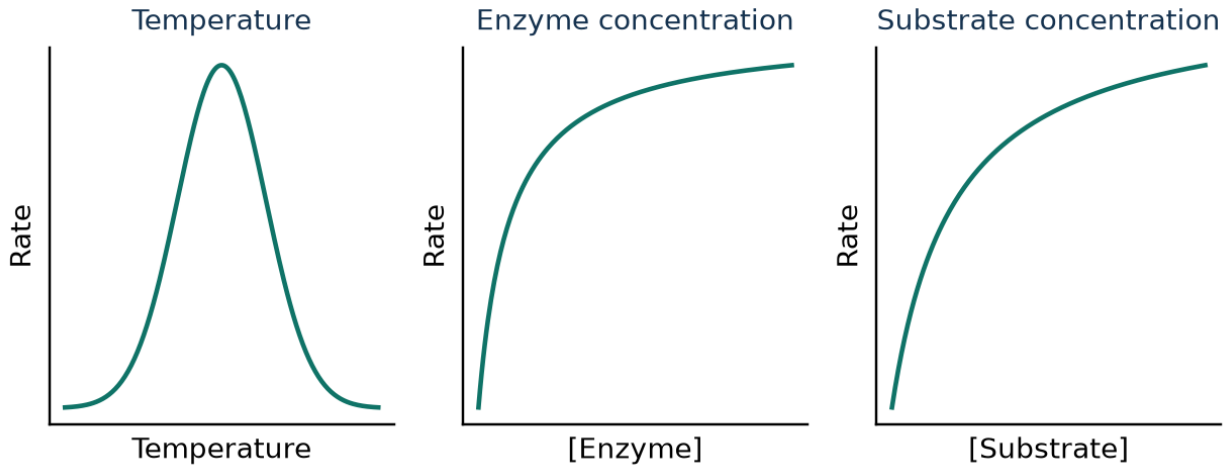


Enzyme-concentration curve rises linearly when substrate is in excess and plateaus when substrate is limiting.

8. Explain the effect of substrate concentration on enzyme rate.

At low substrate concentration, rate rises steeply as more active sites become occupied. As substrate increases, the rise slows because active sites approach saturation.

At high substrate concentration, all active sites are occupied and rate reaches Vmax; adding more substrate has little effect unless more enzyme is added. This gives a hyperbolic saturation curve.



Substrate-concentration curve approaches Vmax as enzymes become saturated.

9. Describe enzymatic inhibition, its types and significance.

Inhibition is reduction of enzyme activity by a molecule called an inhibitor.

- **Competitive inhibition:** inhibitor resembles substrate and competes for the active site; can be overcome by increasing substrate. Example: malonate inhibits succinate dehydrogenase.
- **Non-competitive inhibition:** inhibitor binds at an allosteric site and changes enzyme shape; cannot be fully overcome by substrate. Example: heavy metals, cyanide.
- **Irreversible inhibition:** inhibitor permanently inactivates enzyme, e.g., some toxins and drugs.

Significance: metabolic regulation (feedback inhibition), drug action (sulfa drugs, antibiotics), and toxicology (poisons).

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10. Categorize inhibitors into competitive and non-competitive.

Competitive	Non-competitive
Binding site: active site.	**Binding site:** allosteric/other site.
Structure: resembles substrate.	**Structure:** need not resemble substrate.
Effect of more substrate: can overcome inhibition.	**Effect of more substrate:** cannot fully overcome inhibition.
Example: malonate vs succinate dehydrogenase; sulfa drugs vs PABA.	**Example:** heavy metals, cyanide, feedback ATP inhibition.

11. Explain feedback inhibition.

Substrate → enzyme pathway → end product → End product accumulates → End product inhibits early enzyme → Pathway slows/stops

In feedback (end-product) inhibition, the final product of a metabolic pathway acts as an allosteric inhibitor of an enzyme early in the pathway, usually the first committed step. This prevents overproduction and waste of resources.

Example: ATP inhibits phosphofructokinase in glycolysis when energy is abundant; isoleucine inhibits threonine deaminase in its own synthesis pathway.

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12. Classify enzymes on the basis of reactions catalysed.

Class	Reaction	Example
Oxidoreductases	Oxidation-reduction (electron/hydrogen transfer).	Cytochrome oxidase, dehydrogenases.
Transferases	Transfer functional groups between molecules.	Hexokinase, transaminases.
Hydrolases	Hydrolysis (bond breakage by water).	Lipase, amylase, peptidase.
Lyases	Non-hydrolytic addition/removal of groups, often forming double bonds.	Pyruvate decarboxylase.
Isomerases	Intramolecular rearrangement (isomerization).	Hexose isomerase.
Ligases	Join two molecules using ATP energy.	DNA ligase, polymerases.

This six-class system is the standard international classification of enzymes.

13. Give examples of enzymes named according to substrates.

- **Proteases/peptidases:** act on proteins/peptides — pepsin, trypsin, chymotrypsin.
- **Lipases:** act on lipids — pancreatic lipase.
- **Carbohydrases:** act on carbohydrates — amylase (starch), cellulase (cellulose), maltase (maltose), sucrase (sucrose), lactase (lactose).
- **Nucleases:** act on nucleic acids — DNase, RNase.
- **ATPases:** act on ATP — ATP synthase/hydrolyase.

Naming by substrate makes the enzyme's target clear, though some historical names (pepsin, trypsin) do not follow this pattern.

INQUISITIVE QUESTIONS

1. Does physical exercise involve anabolic processes, catabolic processes, or both?

Both. Catabolic processes break down glucose and fats to release ATP for muscle contraction. Anabolic processes repair and build muscle proteins and replenish glycogen after exercise.

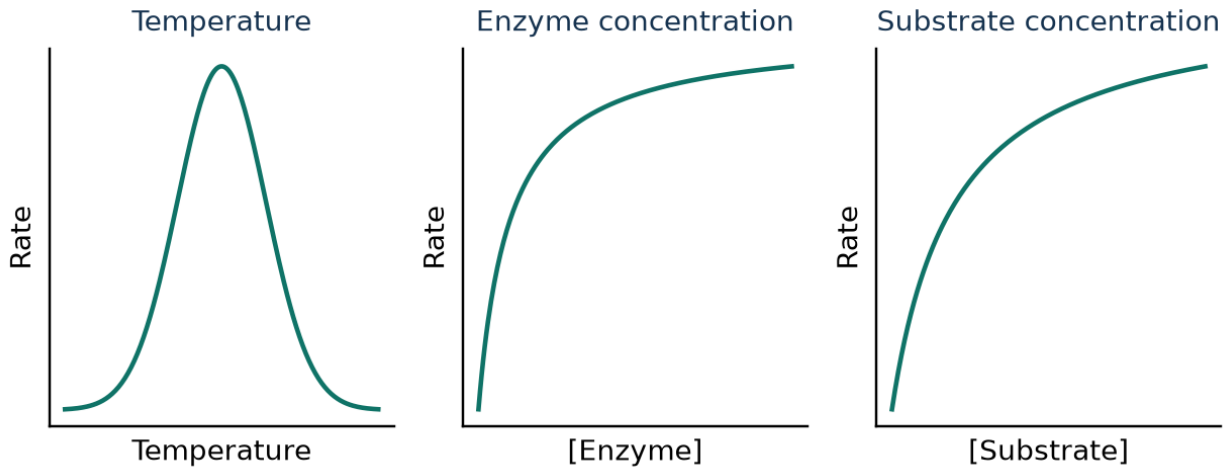
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2. If a reaction can occur without an enzyme, why is an enzyme important?

Without an enzyme the reaction may be too slow to sustain life. Enzymes lower activation energy, allow reactions at mild body conditions, and provide regulation and specificity.

3. Construct and interpret graphs for temperature, enzyme concentration and substrate concentration.

Temperature gives a bell-shaped curve with an optimum; enzyme concentration gives a rising line that plateaus when substrate is limiting; substrate concentration gives a saturation curve reaching Vmax. These graphs show that enzyme activity depends on conditions and availability of active sites.



Interpretation: optimum temperature, linear rise with enzyme when substrate is excess, and saturation with substrate.

4. Identify competitive and non-competitive inhibitors from daily-life chemicals.

Competitive examples include sulfa drugs (compete with PABA in bacteria) and some drugs that mimic substrates. Non-competitive examples include heavy metals (Hg, Pb), cyanide and some pesticides that bind away from the active site and alter enzyme shape.